Geogebra (simulator) condition

Bibliography: Trucios, C., & Trujillo, M. (2021). Chapter 2/ Triangle. In *Plane and spatial geometry* (7th ed., pp. 34–56). essay, Fondo Editorial Rodo.

Theorem # 1 Internal Δ angles sum

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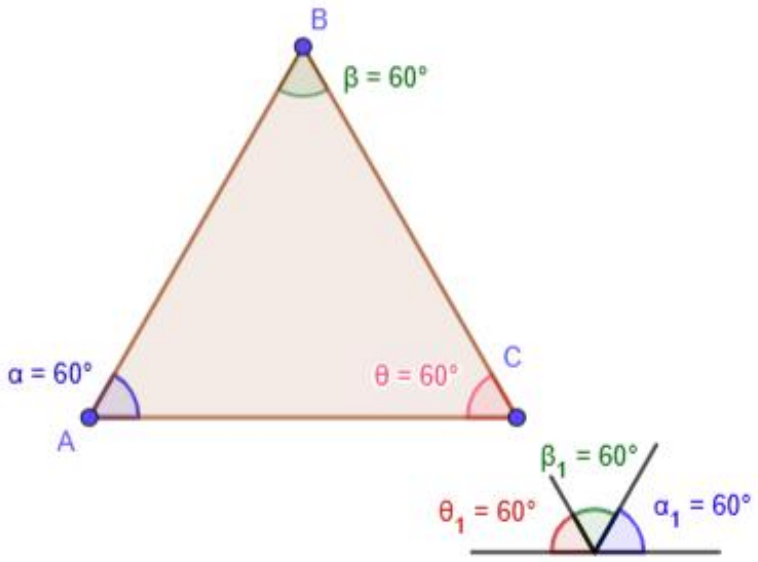
Try dragging points A, B, and C to see how the triangle transforms.

In every triangle, the sum of the measures of the angles is equal to 180°

$$\theta + \alpha + \beta = 180^\circ$$

$$60^\circ + 60^\circ + 60^\circ = 180^\circ$$



Link to activity (Theorem 1): https://www.geogebra.org/m/fmqpszh9

Theorem # 2 External Δ angles sum

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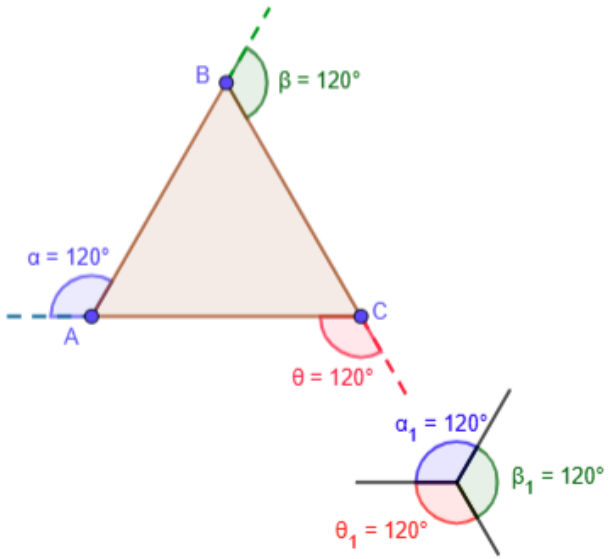
Try dragging points A, B, and C to see how the triangle transforms.

In any triangle, the sum of the exterior angles, one at each vertex, is 360°.

$$120^\circ + 120^\circ + 120^\circ = 360^\circ$$

$$\alpha + \beta + \theta = 360^\circ$$



Link to activity (Theorem 2): https://www.geogebra.org/m/a79r98tg

Theorem # 3 Exterior angle

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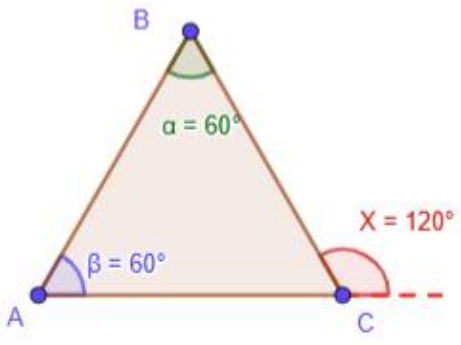
Try dragging points A, B, and C to see how the triangle transforms.

In every triangle, the measure of an exterior angle
is equal to the sum of the measures of the two
interior angles that are not adjacent to that exterior angle.

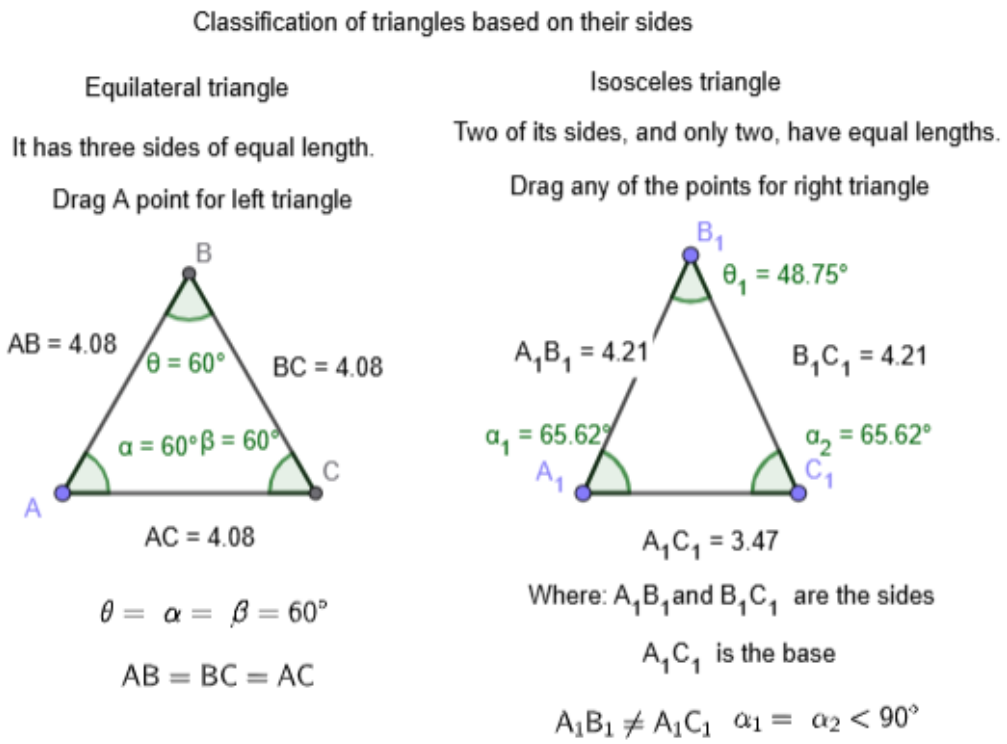
$$120^\circ = 60^\circ + 60^\circ$$

$$X = \alpha + \beta$$



Link to activity (Theorem 3): https://www.geogebra.org/m/w4nfxt4q

Classification of triangles based on their sides



Link to activity (Classification of triangles based on their sides): https://www.geogebra.org/m/h66hc9ee